

Lei de TRANSFORMAÇÃO DA CONEXÃO

$$q^i \rightarrow q^{i'}$$

$$\Gamma^i_{jk} \rightarrow \Gamma^{i'}_{j'k'}$$

Lembrando que:

$$V^k_{;j} = V^k_{,j} + \Gamma^k_{mi} V^m \longrightarrow V^{k'}_{;j'} = V^{k'}_{,j'} + \Gamma^{k'}_{m'i'} V^{m'}$$

e Sabemos que sob $q^i \rightarrow q^{i'}$ V se transforma $V^{k'}_{;j'} = \frac{\partial q^k}{\partial q^{k'}} \frac{\partial q^l}{\partial q^{j'}} V^r_{;l}$

Desta forma:

$$V^{k'}_{,j'} + \Gamma^{k'}_{m'i'} V^{m'} = \frac{\partial q^k}{\partial q^{k'}} \frac{\partial q^m}{\partial q^{j'}} (V^m_{,n} + \Gamma^m_{rn} V^r)$$

$$\frac{\partial V^{k'}}{\partial q^{j'}} + \Gamma^{k'}_{m'i'} V^{m'} = \frac{\partial q^k}{\partial q^{k'}} \frac{\partial q^m}{\partial q^{j'}} \left(\frac{\partial V}{\partial q^n} + \Gamma^m_{rn} V^r \right) \quad (1)$$

passando $q^i \rightarrow q^i$

$$\frac{\partial V^{k'}}{\partial q^{j'}} = \frac{\partial}{\partial q^{j'}} \left(\frac{\partial q^k}{\partial q^{k'}} V^m \right) = \frac{\partial^2 q^k}{\partial q^{i'} \partial q^{j'}} V^m + \frac{\partial q^k}{\partial q^{k'}} \frac{\partial V^m}{\partial q^{j'}} \quad (2)$$

$$= \frac{\partial^2 q^k}{\partial q^{j''} \partial q^{j'}} \frac{\partial q^m}{\partial q^{i'}} V^m + \frac{\partial q^k}{\partial q^{k'}} \frac{\partial V^m}{\partial q^r} \frac{\partial q^r}{\partial q^{i'}} \quad (3)$$

(3) $\rightarrow$ (1)

$$\frac{\partial^2 q^k}{\partial q^m \partial q^n} \frac{\partial q^m}{\partial q^i} v^m + \frac{\partial^2 q^k}{\partial q^m \partial q^n} \frac{\partial q^m}{\partial q^i} \frac{\partial q^r}{\partial q^i} + \Gamma^{k'}_{m'i} v^{m'} = \frac{\partial q^k}{\partial q^m} \frac{\partial q^m}{\partial q^i} \left(\frac{\partial v}{\partial q^m} + \Gamma^m_{rm} v^r \right) \quad (1)$$

$v^{m'} = \frac{\partial q^{m'}}{\partial q^i} v^i$

$$\frac{\partial^2 q^k}{\partial q^m \partial q^n} \frac{\partial q^m}{\partial q^i} v^m + \cancel{\frac{\partial^2 q^k}{\partial q^m \partial q^n} \frac{\partial q^m}{\partial q^i} \frac{\partial q^r}{\partial q^i}} + \Gamma^{k'}_{m'i} \frac{\partial q^{m'}}{\partial q^i} v^i = \cancel{\frac{\partial q^k}{\partial q^m} \frac{\partial q^m}{\partial q^i} \frac{\partial v}{\partial q^m}} + \frac{\partial q^k}{\partial q^m} \frac{\partial q^m}{\partial q^i} \Gamma^m_{rm} v^r \quad (1)$$

$$\cancel{\frac{\partial^2 q^k}{\partial q^m \partial q^n} \frac{\partial q^m}{\partial q^i} v^i} + \Gamma^{k'}_{m'i} \cancel{\frac{\partial q^{m'}}{\partial q^i} v^i} = \frac{\partial q^k}{\partial q^m} \frac{\partial q^m}{\partial q^i} \Gamma^m_{rm} v^r$$

$$\frac{\partial^2 q^k}{\partial q^m \partial q^n} \frac{\partial q^m}{\partial q^i} \frac{\partial q^r}{\partial q^i} + \Gamma^{k'}_{m'i} \frac{\partial q^{m'}}{\partial q^i} \frac{\partial q^r}{\partial q^i} = \frac{\partial q^k}{\partial q^m} \frac{\partial q^m}{\partial q^i} \frac{\partial q^r}{\partial q^i} \Gamma^m_{rm}$$

$\underbrace{\frac{\partial q^{m'}}{\partial q^i} \frac{\partial q^r}{\partial q^i}}_{\delta^{m'}_i} \underbrace{\Gamma^{k'}_{m'i}}_{\Gamma^{k'}_{j'i}}$

Assim:

A conexão $\tilde{\Gamma}$ é um Tensor

$$\Gamma^{k'}_{j'i} = \Gamma^m_{rm} \frac{\partial q^r}{\partial q^j} \frac{\partial q^m}{\partial q^i} \frac{\partial q^k}{\partial q^m} - \frac{\partial^2 q^k}{\partial q^m \partial q^n} \frac{\partial q^m}{\partial q^i} \frac{\partial q^r}{\partial q^i}$$

↳ CONEXÃO AFIM ou afinidade

Derivada Covariante de um tensor:

$$T^{i'o} = U^i v^i$$